

CU Online — Chandigarh University

MCA Final Year Project — Milestone Report

MILESTONE 6 REPORT

Document Multi-Cloud Cost Strategy

Final strategy documentation, savings projections, and project completion summary

Project Title: Multi-Cloud Cost Comparison and Optimization Analytics Dashboard

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Program: Master of Computer Applications (MCA) | CU Online

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Milestone 6: Document Multi-Cloud Cost Strategy

The final milestone of the project brings together all findings from the preceding milestones into a comprehensive, documented multi-cloud deployment strategy. This milestone encompasses the formal statement of the recommended provider allocation, the quantification of projected savings relative to alternative strategies, the validation of project objectives achievement, and the documentation of the complete project as submitted for academic evaluation.

6.1 Recommended Multi-Cloud Strategy

Based on the five-year cost trend analysis, the workload-to-provider efficiency mapping from Milestone 5, and the structural discount mechanism analysis from Milestone 1, the recommended multi-cloud deployment strategy is a three-provider allocation with the following workload distribution:

Provider	Allocation	Primary Workloads	Discount Mechanism	Cost Advantage
GCP	40%	Analytics, batch processing, data pipelines, ML training, BigQuery workloads	Sustained-use auto-discount (no commitment); committed-use for steady-state	27–30% cheaper than AWS; 46% cost/user improvement FY2021–FY2025
AWS	40%	Production APIs, managed services (Lambda, Aurora, CloudFront), global CDN	Reserved Instances (1–3yr) for baseline; on-demand for variable	Necessary for managed service breadth; Reserved Instances: 40–72% discount
Azure	20%	Enterprise identity (Entra ID), Active Directory, M365 integration, Windows Server VMs	Hybrid Benefit for existing Windows/SQL licenses (40–60% reduction)	~15% saving vs AWS for Microsoft-licensed workloads; best-in-class AD integration

Table M6.1: Recommended Multi-Cloud Deployment Strategy — Provider Allocation

6.2 Projected Savings Analysis

The financial impact of the recommended multi-cloud strategy versus alternative single-provider approaches was quantified using the five-year cost data and the workload-specific pricing analyses. All projections assume FY2025 workload scale as the baseline.

Strategy	Annual Cost Estimate (\$M)	5-Year Cumulative (\$M)	Saving vs 100% AWS	Assessment
100% AWS	~\$328M	~\$1,244M	Baseline (0%)	Maximum cost; vendor lock-in; Reserved Instances partially mitigate
100% Azure	~\$291M	~\$1,056M	~11.3%	Limited — Hybrid Benefit applies only to Microsoft workloads
100% GCP	~\$238M	~\$830M	~27.4%	Good for analytics-heavy org; loses AWS managed service breadth
60% AWS / 40% GCP	~\$278M	~\$1,034M	~15%	Better; loses Azure identity advantage
Recommended: 40% GCP / 40% AWS / 20% Azure	~\$260–270M	~\$960–995M	~18–22%	OPTIMAL — balances all three discount mechanisms and service strengths

Table M6.2: Cost Strategy Comparison — Projected Savings vs 100% AWS Approach

6.3 Project Objectives Achievement Verification

Milestone	Objective	Deliverable Produced	Status
M1	Study pricing models of AWS, Azure, GCP	Pricing model documentation; comparative analysis table; simulation parameters	Completed ✓
M2	Simulate infrastructure costs on each platform	60-record quarterly dataset; 5-sheet Excel workbook structure	Completed ✓
M3	Store cost comparison data	Power BI data model; Power Query pipeline; 2 bugs resolved (BUG-001, BUG-002)	Completed ✓
M4	Build visualization dashboard	9-visual interactive Power BI dashboard; 2 pages; 4 DAX measures; slicers; drill-through	Completed ✓
M5	Identify cheapest deployment options	Workload-to-provider mapping; cost-per-user trend analysis; provider comparison	Completed ✓
M6	Document multi-cloud cost strategy	This document + Recommendations sheet + Chapter 7/8 of project report	Completed ✓

Table M6.3: Project Objectives Achievement Verification

6.4 Core Findings Summary

- GCP is 27–30% cheaper than AWS in every financial year from FY2021 to FY2025 — the cost advantage is structural and persistent, not cyclical.
- GCP's cost-per-user improved 46% over five years (from \$35.86K to \$18.67K), outpacing Azure's 37% and AWS's 20% improvement — GCP's efficiency advantage is accelerating.
- The five-year cumulative cost saving from using GCP instead of AWS for equivalent workloads is \$100.4M — a figure that scales linearly with cloud spend volume.
- The recommended 40% GCP / 40% AWS / 20% Azure multi-cloud allocation delivers 18 to 22 percent total cost reduction versus a 100% AWS strategy, representing \$28M to \$34M in annual savings at FY2025 workload scale.
- All 10 non-functional requirements were met or exceeded, including sub-second average slicer response (1.4 seconds vs 2-second target) and full dataset refresh in 6.1 seconds vs 15-second target.
- The project demonstrates that a production-quality multi-cloud analytics dashboard can be built at zero cost using Power BI Desktop and Excel — fully reproducible from the GitHub repository.

6.5 Project Artefacts Published to GitHub

- README.md — complete project overview, installation instructions, and key findings summary.
- Multi_Cloud_Cost_Dashboard_Updated.xlsx — source dataset with all 5 sheets.
- Multi_Cloud_Cost_Dashboard_Vishal.pbix — complete Power BI dashboard.
- docs/Project_Report_Vishal_Kumar_MCA.pdf — full project report.
- docs/Presentation_Vishal_Kumar_MCA.pptx — 22-slide project presentation.
- docs/PowerBI_Dashboard_Guide.docx — step-by-step dashboard reproduction guide.
- milestones/ — all 6 milestone reports (Milestones 1 through 6).